IA ASSIGNMENT 1

Project Management

Q.1) GreenVille Road Project

Given activities:

Activity	Description	Duration	Predecessor
A	Land survey	5 days	None
B	Design	10 days	A
C	Permits	7 days	A
D	Material procurement	12 days	B
E	Construction	30 days	C, D
F	Inspection	4 days	E

(a) Network diagram

Start → A (Land survey, 5 days) → B (Design, 10 days) → D (Material procurement, 12 days) → E (Construction, 30 days) → F (Inspection, 4 days) → Finish

A also leads to C (Permits, 7 days). Both C and D must be completed before E can start.

(b) Critical path and project duration

Path 1: A → B → D → E → F = 5 + 10 + 12 + 30 + 4 = 61 days.

Path 2: A → C → E → F = 5 + 7 + 30 + 4 = 46 days.

The longest path is A → B → D → E → F. Therefore, the critical path is A-B-D-E-F and the project duration is 61 days.

(c) Float (slack) for each non-critical activity

Total Float = Latest Start − Earliest Start = Latest Finish − Earliest Finish.

Activity	ES	EF	LS	LF	Float
A	0	5	0	5	0
B	5	15	5	15	0
C	5	12	20	27	15 days
D	15	27	15	27	0
E	27	57	27	57	0
F	57	61	57	61	0

Only activity C is non-critical. Its float is 20 − 5 = 15 days. All other activities have zero float.

(d) Effect of a 5-day permit delay

No, the completion date will not change. Activity C has 15 days of float, and a 5-day delay uses only 5 days of it. The remaining float is 15 − 5 = 10 days. The project still finishes in 61 days, provided there are no other delays.

Q.2) EcoBuild – Project Management

(a) SMART project goal

SMART means Specific, Measurable, Achievable, Relevant and Time-bound.

Goal: To design and construct a 10-storey eco-friendly office complex within 18 months and a budget of 25 crore, using energy-efficient lighting, water-saving systems and sustainable materials, while meeting all applicable safety and building regulations and reducing estimated energy consumption by at least 20% compared with a similar conventional building.

Note: The budget, timeline and 20% target are illustrative assumptions because they are not provided in the question. They should be confirmed with the client.

(b) Four project management knowledge areas

Knowledge area	Explanation	Example
Scope management	Defines the work included in the project.	Decide floors, offices, parking and green features.
Time management	Plans and controls the project schedule.	Schedule design, foundation, construction and inspection.
Cost management	Estimates and controls expenses.	Monitor materials, labour, equipment and solar costs.
Quality management	Ensures work meets required standards.	Check construction quality, safety and energy efficiency.

(c) Project manager responsibilities

Planning phase:

1. Define the project scope and deliverables.
2. Prepare the schedule, activities, milestones and deadlines.
3. Estimate costs and prepare the project budget.
4. Allocate people, equipment and materials.
5. Identify risks such as weather, shortages, cost increases and safety issues.
6. Plan quality standards and stakeholder communication.

Execution phase:

1. Coordinate engineers, contractors and workers.
2. Monitor progress against the schedule.
3. Control spending against the approved budget.
4. Ensure construction quality and workplace safety.
5. Manage changes through the agreed approval process.
6. Communicate updates and resolve problems.
7. Take corrective action when delays or unexpected issues occur.

Q.3) Change Request and Team Conflict

(a) Steps to handle the change request

1. Record the request: Document the client's request for rooftop solar panels, including capacity and expected benefits.

2. Analyse the impact: Check structural strength, electrical design, safety, cost and installation time.

3. Consult stakeholders: Discuss the findings with both engineers, the client, contractor and relevant specialists.

4. Obtain approval: Submit the impact analysis to the authorised client or change-control board for approval or rejection.

5. Update the project plan: If approved, update the scope, schedule, budget, drawings, resources and risk register.

6. Implement and monitor: Assign responsibilities, install the system according to the approved design and inspect the work.

(b) Resolving the team conflict

1. Listen to both engineers and understand their reasons.

2. Arrange a meeting where both can explain their technical views.

3. Review evidence such as structural calculations, electrical requirements, safety standards and drawings.

4. Consult a structural engineer or solar specialist if required.

5. Make a decision based on safety, technical facts and project needs.

6. Explain the decision, assign responsibilities and document the outcome.

Example: One engineer believes the roof can support the panels, while another believes reinforcement is needed. The project manager should obtain a structural assessment before installation.

(c) Effects on scope, time and cost

Constraint	Possible effect	Action
Scope	Adds panels, mounting structures, wiring and testing.	Update scope statement and WBS.
Time	Design, procurement, installation and inspection may take extra time.	Revise schedule, milestones and critical path.
Cost	Equipment, labour, design and roof reinforcement may add cost.	Revise budget and seek approval for extra funds.

The project manager must update technical drawings, schedule, cost baseline after approval, resource allocation, risk register and safety plan. The approved changes must be communicated to all stakeholders and monitored against the revised plan.

Q.4) E-commerce Website Development

(a) Work Breakdown Structure (WBS) up to three levels

Level 1: 1.0 E-commerce Website Development

1.1 Requirements

- *1.1.1 Gather user requirements*
- *1.1.2 Analyse functional requirements*
- *1.1.3 Finalise specifications*

1.2 Design

- *1.2.1 Design UI/UX*
- *1.2.2 Design database*
- *1.2.3 Design architecture*

1.3 Development

- *1.3.1 Develop frontend*
- *1.3.2 Develop backend and APIs*
- *1.3.3 Integrate cart and payment*

1.4 Testing

- *1.4.1 Perform unit testing*
- *1.4.2 Perform integration testing*
- *1.4.3 Perform system testing*

1.5 Deployment

- *1.5.1 Configure production server*
- *1.5.2 Deploy website*
- *1.5.3 Verify and hand over*

(b) Gantt chart for 12 weeks (3 months)

Activity	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12
Requirements	■	■										
Design		■	■	■								
Development				■	■	■	■	■	■			
Testing								■	■	■	■	
Deployment											■	■

The schedule is illustrative. Requirements take Weeks 1–2, design Weeks 2–4, development Weeks 4–9, testing Weeks 8–11, and deployment Weeks 11–12.

(c) Risks of overlapping testing and development, and management

Risks:

1. Incomplete features may produce misleading test failures.

2. Code changes may require tests to be repeated.

3. Modules may fail when integrated, even if they work separately.

4. Poor communication may cause misunderstandings and unresolved defects.

5. Many defects discovered late can delay deployment.

How to manage them:

1. Test each completed module incrementally.

2. Define clear entry and exit criteria for testing.

3. Use continuous integration and automated tests.

4. Track and prioritise bugs using a defect-tracking tool.

5. Hold regular discussions between developers and testers.

6. Reserve enough time for final integration, security, performance and user acceptance testing.

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